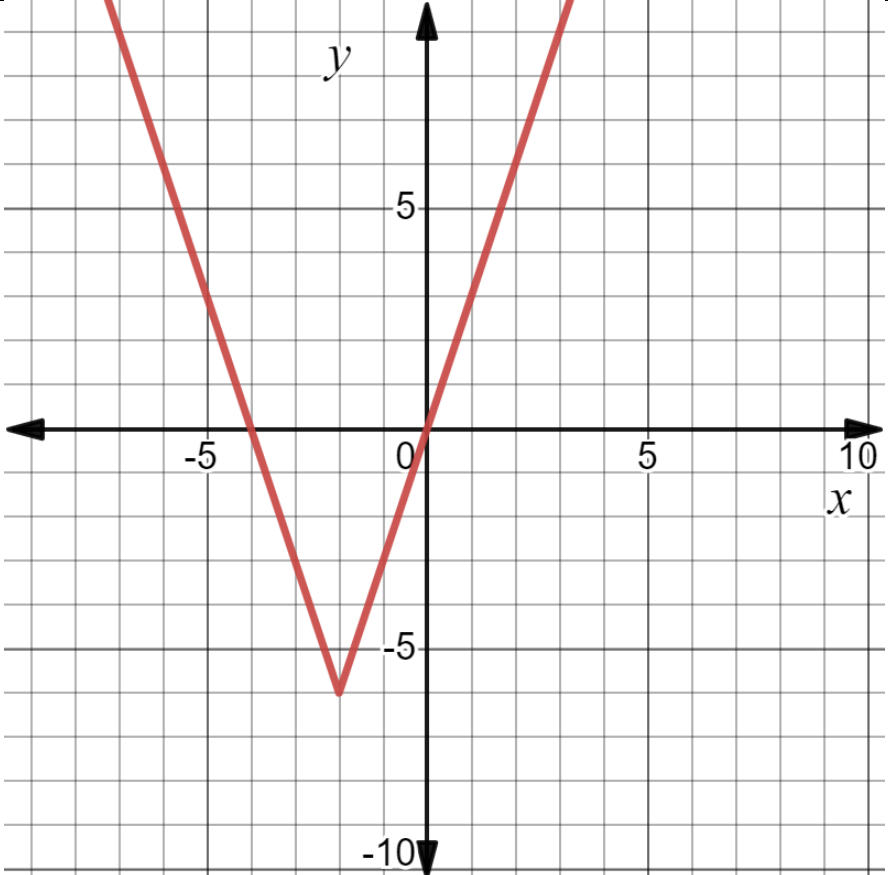


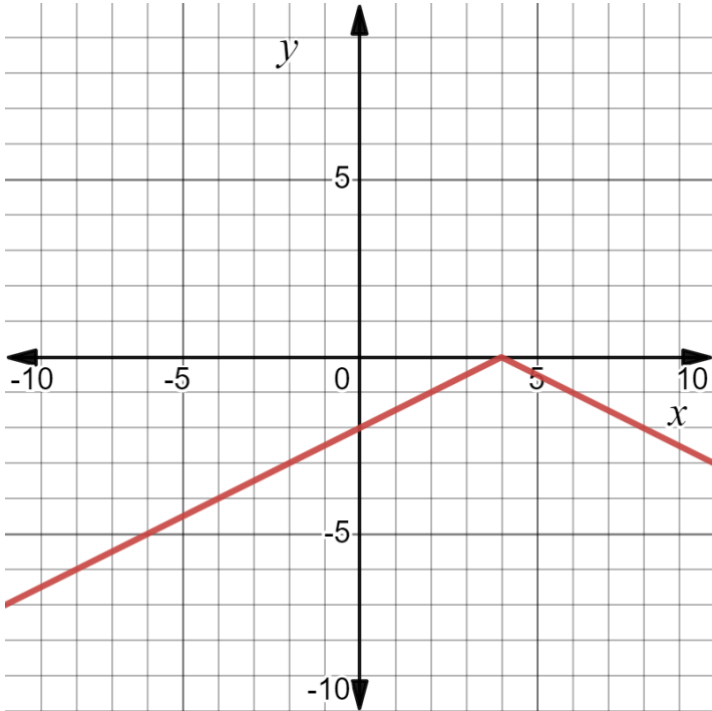
Algebra 1 Honors Unit 10 Assessment Guide

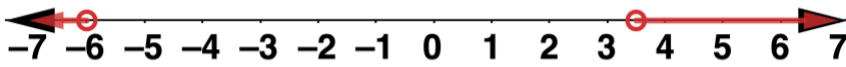
General Information	Unit 10 extends students' understanding of functions to the absolute value function. The unit supports students' understanding of the different types of solutions using graphs. Students may connect these types of solutions to solutions of quadratic functions which helps support the spiral learning of concepts found in the Algebra 1 benchmarks. Similarly, students may notice that the key features for absolute value functions are similar to key features of quadratic functions. Also, solving absolute value equations is connected to solving linear equations. Teachers can use these connections to establish where students may struggle on concepts that were previously learned. As teachers work with small groups or individuals to support struggling learners, teachers are encouraged to make connections with prior learning. Research shows that having students recall content after a spaced time helps students better remember the content. This type of spiraling of the content can also help as the end of the year approaches. Honors students also solve absolute value inequalities. Students are introduced to the concept by looking at solutions on a number line and discovering how the solution set on the number line is connected to absolute value inequalities. Students may confuse real-world applications of absolute value equations and real-world applications of absolute value inequalities. Teachers can help students distinguish between the two by determining if all values within a tolerance are to be determined or if just the maximum and minimum are to be determined.
Calculator Information	The questions on the assessment are calculator neutral. Teachers can determine whether to allow students the use of a calculator.
Total Recommended Points	This assessment is a total of 100 points.

Question	1	Benchmark(s)	MA.912.AR.4.1
	2 and 32	Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 2 points for set-up • 2 points for correct work deducting points for minor errors • 2 points for the correct solutions
Question	2	Benchmark(s)	MA.912.AR.4.3
		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	3	Benchmark(s)	MA.912.AR.4.1
	–6 and 9	Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 4 points for correct work deducting points for minor errors • 2 points for the correct solutions
Question	4	Benchmark(s)	MA.912.AR.4.1
	No solutions	Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 4 points for correct work deducting points for minor errors • 2 points for the correct solutions Students who notice that the answer is no solution before showing work should earn full credit.
Question	5	Benchmark(s)	MA.912.AR.4.1
	13 and –25	Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 4 points for correct work deducting points for minor errors • 2 points for the correct solutions

Question	6	Benchmark(s)	MA.912.AR.4.1
	4 and -14	Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 4 points for correct work deducting points for minor errors • 2 points for the correct solutions
Question	7	Benchmark(s)	MA.912.AR.4.1
	$ x - 691.15 = 3.15$	Recommended Scoring (4 Total Points)	No partial credit is suggested.

Question		8a		Benchmark(s)	MA.912.AR.4.3																								
<table><tr><th>Name</th><th>Key Feature</th><th>Name</th><th>Key Feature</th></tr><tr><td>x-intercepts</td><td>$(4, 0)$</td><td>increasing</td><td>$x < 4$</td></tr><tr><td>y-intercept</td><td>$(0, -2)$</td><td>decreasing</td><td>$x > 4$</td></tr><tr><td>vertex</td><td>$(4, 0)$</td><td>positive</td><td>none</td></tr><tr><td>axis of symmetry</td><td>$x = 4$</td><td>negative</td><td>$x < 4$ or $x > 4$</td></tr><tr><td>domain</td><td>$\mathbb{R}$</td><td>range</td><td>$-\infty < y \leq 0$</td></tr></table>				Name	Key Feature	Name	Key Feature	x -intercepts	$(4, 0)$	increasing	$x < 4$	y -intercept	$(0, -2)$	decreasing	$x > 4$	vertex	$(4, 0)$	positive	none	axis of symmetry	$x = 4$	negative	$x < 4$ or $x > 4$	domain	$\mathbb{R}$	range	$-\infty < y \leq 0$	Recommended Scoring (30 Total Points)	Consider giving students 3 points for each key feature.
Name	Key Feature	Name	Key Feature																										
x -intercepts	$(4, 0)$	increasing	$x < 4$																										
y -intercept	$(0, -2)$	decreasing	$x > 4$																										
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Question		8b		Benchmark(s)	MA.912.AR.4.3																								
				Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.																								

Question	9	Benchmark(s)	MA.912.AR.4.2
$x > 3.5 \quad x < -6$ 		Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 2 points for correct work deducting points for minor errors • 2 points for the correct solutions • 2 points for the correct graph
Question	10	Benchmark(s)	MA.912.AR.4.2
$2.5 \leq x \leq 7.5$ 		Recommended Scoring (6 Total Points)	Consider giving partial credit using the following: • 2 points for correct work deducting points for minor errors • 2 points for the correct solutions • 2 points for the correct graph
Question	11a	Benchmark(s)	MA.912.AR.4.2
$ x - 2.09 \leq 0.88$		Recommended Scoring (6 Total Points)	No partial credit is suggested.
Question	11b	Benchmark(s)	MA.912.AR.4.2
		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.